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Off-Time Puberty in Autism Spectrum Disorder and ADHD: A Scoping Review of Pubertal Timing, Social Cognition, and Psychopathological Risk

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Abstract

Off-time pubertal development defined as pubertal onset or progression deviating meaningfully from age- and sex-matched normative expectations confers heightened risk for internalizing and externalizing psychopathology during adolescence. Children and adolescents with neurodevelopmental disorders (NDDs), including autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD), carry pre-existing impairments in theory of mind (ToM) the capacity to represent and respond to the mental states of others that substantially compromise their capacity to navigate the increasingly complex social demands that puberty introduces. Evidence indicates that ASD is associated with elevated rates of precocious puberty and altered pubertal tempo, and that ADHD in females is linked to earlier pubertal onset with consequent internalizing symptom exacerbation; yet the mechanisms connecting pubertal timing, ToM impairment, and psychopathological risk in these populations have not been systematically reviewed. This scoping review, guided by the PRISMA extension for Scoping Reviews (PRISMA-ScR), synthesises evidence from 19 studies (2016–2026) examining pubertal timing in ASD and ADHD, the neurobiological mechanisms linking puberty to social cognition, and the clinical and psychopathological consequences of their convergence. We identify three mechanisms through which off-time puberty and pre-existing ToM impairment interact: (1) demand-capacity mismatch amplification, (2) masking and camouflaging under elevated social load, and (3) physical-cognitive developmental asynchrony. No primary studies have directly examined the moderating role of ToM on the puberty-psychopathology relationship in NDD populations, representing a critical research gap. We propose a dual-vulnerability framework and outline research priorities for future longitudinal and intervention studies in this underserved population.

Keywords Off-time puberty · Neurodevelopmental disorders · Autism spectrum disorder · ADHD · Theory of mind · Social cognition · Pubertal timing · Masking · Psychopathology · Scoping review

Introduction

The onset of puberty represents one of the most consequential biological transitions of the lifespan. Beyond well-documented physical changes, puberty initiates cascading effects on brain development, social cognition, and peer relationship complexity that fundamentally reshape adolescent mental health vulnerability (Pfeifer & Allen, 2021). Pubertal timing defined as the relative onset of puberty compared to same-aged, same-sex peers has emerged as a key moderator of psychopathological risk during this period, with earlier timing in females consistently associated with internalizing psychopathology, peer victimisation, and social exclusion (Graber, 2013; Mendle et al., 2010). These effects are multiply determined, encompassing psychosocial stressors related to maturational deviance, neurobiological changes in limbic-prefrontal circuitry, and hormonal modulation of mood regulation and social salience systems (Goddings et al., 2012; Morningstar & Burns, 2025).

A critical and systematically overlooked feature of this literature is its near-exclusive focus on neurotypically developing populations. Children and adolescents with autism

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spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD) represent a substantial proportion of the youth population with ASD affecting approximately 1 in 36 children (Maenner et al., 2023) and ADHD approximately 7.2% globally (Polanczyk et al., 2014). Both conditions are characterised by pre-existing impairments in theory of mind (ToM): the capacity to represent and respond to the mental states of others. In ASD, both cognitive and affective ToM are affected; in ADHD, deficits concentrate in affective ToM and social perspective-taking under attentional load, while cognitive ToM is largely preserved (Bolte, 2025; Capuozzo et al., 2024; Haza et al., 2024). ToM underpins the social navigation skills reading social hierarchies, interpreting peer feedback, managing romantic interest, and maintaining friendships that become critically important during adolescence. When puberty arrives off-time for a child whose ToM is already substantially impaired, the developmental consequences are not simply additive: an accelerating social environment meets a social-cognitive system that was already operating below normative capacity.

Despite the theoretical and clinical significance of this intersection, no scoping review has systematically examined the evidence for altered pubertal timing in ASD and ADHD, the neurobiological mechanisms linking puberty to social cognition, and the consequences of their convergence for psychopathological outcomes. The present scoping review addresses this gap, with the following objectives: (1) to synthesise evidence on pubertal timing in ASD and ADHD relative to typically developing peers; (2) to examine the neurobiological mechanisms through which pubertal hormonal changes modulate social-cognitive development in these populations; (3) to map the psychopathological consequences of the convergence of off-time pubertal timing and pre-existing ToM impairment; and (4) to identify research and clinical priorities arising from this synthesis.

Method

Design and Protocol

This review was conducted in accordance with the PRISMA extension for Scoping Reviews (PRISMA-ScR; Tricco et al., 2018) and the methodological framework for scoping reviews proposed by Arksey and O'Malley (2005) as refined by Levac et al. (2010). A protocol was not pre-registered, consistent with the exploratory nature of the review.

Search Strategy

Systematic searches were conducted in PubMed/MEDLINE, PsycINFO, Web of Science, and Scopus in May 2026. The search strategy combined three conceptual domains using Boolean operators: (1) pubertal timing: ("pubertal timing" OR "off-time puberty" OR "early puberty" OR "precocious puberty" OR "late puberty" OR "delayed puberty" OR "pubertal tempo" OR "adrenarche" OR "menarche" OR "Tanner stage"); (2) neurodevelopmental disorders: ("autism spectrum disorder" OR "ASD" OR "autistic" OR "attention-deficit/hyperactivity disorder" OR "ADHD" OR "attention deficit"); (3) outcomes: ("theory of mind" OR "social cognition" OR "mentalizing" OR "psychopathology" OR "internalizing" OR "externalizing" OR "anxiety" OR "depression" OR "camouflaging" OR "masking"). Searches were limited to English-language publications from January 2010 to April 2026. Reference lists of included articles and relevant systematic reviews were hand-searched for additional eligible studies.

Eligibility Criteria

Studies were eligible for inclusion if they: (1) examined pubertal timing, pubertal tempo, or hormonal correlates of puberty in children or adolescents with ASD or ADHD; OR examined the relationship between pubertal development and social cognition, ToM, or social-cognitive brain development in ASD, ADHD, or typically developing youth relevant to NDD populations; OR examined masking/camouflaging and mental health outcomes during adolescence in ASD or ADHD; (2) included participants aged 8–18 years; (3) were published in peer-reviewed journals. Studies were excluded if they: (1) focused exclusively on adult populations; (2) examined puberty only in the context of medication side effects without examining pubertal timing per se; (3) did not include a clinical or population sample with ASD or ADHD (for primary studies); or (4) were case reports.

Data Extraction and Quality Assessment

Methodological quality of included primary studies was assessed using the JBI Critical Appraisal Checklist for Prevalence Studies and the JBI Critical Appraisal Checklist for Systematic Reviews, adapted to the study designs included (Moola et al., 2020). Each study was rated on eight criteria relevant to its design: (1) sampling representativeness; (2) validity and reliability of puberty measurement; (3) presence of a comparison group; (4) control for key confounders (age, sex, IQ, ethnicity); (5) sex-stratified analyses; (6) appropriate statistical methods; (7) adequate follow-up (for longitudinal studies); and (8) transparency of reporting. Studies were rated as high ($\geq$7/8), moderate-high (5–6/8), or moderate (3–4/8) quality. Narrative synthesis was used to integrate findings, given the heterogeneity in study designs, populations, and outcome measures.

Results

Search Results and Study Characteristics

The initial database searches identified 847 records (PubMed: 312, PsycINFO: 218, Web of Science: 196, Scopus: 121). After removal of duplicates ($n = 203$), 644 records were screened by title and abstract. Following application of eligibility criteria, 54 full-text articles were assessed, of which 19 met inclusion criteria (see Fig. 1 for PRISMA-ScR flow diagram). Of the 19 included studies, 9 examined pubertal timing directly in ASD or ADHD populations (7 ASD; 4 ADHD; 2 overlapping), 3 examined hormonal mechanisms linking puberty to social-cognitive development, 3 examined masking and camouflaging at pubertal transition, and 4 examined ToM development across adolescence (1 in ASD, 1 in ADHD, 2 in typically developing samples providing normative context). Methodological quality ranged from moderate to high across included studies (Table 1). The most significant methodological limitation across the literature was the absence of any primary study that directly examined the moderating role of ToM capacity on the association between pubertal timing and psychopathological outcomes in ASD or ADHD populations.

Methodological quality ratings for all included primary studies are presented in Table 1.

Pubertal Timing in ASD

Primary Studies: Physical Examination-Based Evidence

Six primary studies examined pubertal timing in ASD using direct physical assessment (Tanner staging) or biomarkers, yielding a consistent pattern of altered pubertal development relative to typically developing (TD) peers. Corbett and colleagues conducted the most methodologically rigorous longitudinal programme of research in this area. In a cross-sectional study of 239 youth (137 ASD, 102 TD) aged 10 to 13 years, females with ASD showed significantly earlier menarche ($t = -2.56$, $p = .018$) and advanced Tanner staging relative to TD females, while no significant timing differences were observed in males (Corbett et al., 2020). A subsequent three-wave longitudinal study of 244 youth confirmed earlier pubertal onset in both sexes with ASD and demonstrated faster pubertal tempo in males, with ASD males showing a significantly accelerated progression through Tanner stages relative to TD males (Corbett et al., 2022). The same longitudinal cohort was further characterised hormonally: ASD youth showed significantly elevated morning salivary testosterone concentrations relative to TD peers, with a significant age-by-sex interaction revealing distinct developmental slopes (Muscatello et al., 2022). Most recently, a four-wave longitudinal analysis spanning ages 10 to 17 years demonstrated that testosterone and cortisol followed a coupled developmental trajectory in ASD males, with hormone levels mediating the relationship between diagnostic status and behavioural outcomes across puberty (Corbett et al., 2025).

Population-Level Epidemiological Evidence

Two large-scale epidemiological studies provide important complementary evidence using clinical diagnoses of precocious puberty (PP) rather than Tanner staging, permitting examination of risk at the population level. Geier and Geier (2021), using Florida Medicaid data ($N = 101{,}736$), found that the incidence of PP was 3.15 times higher in ASD youth relative to undiagnosed peers, with a Cox proportional hazard ratio of 4.64 (95% CI: 3.11–6.93). Tsai and colleagues (2024), using Taiwan's National Health Insurance Research Database ($N = 22{,}208$), reported that ASD was an independent risk factor for PP regardless of comorbid intellectual disability, with a hazard ratio of 6.48 for autistic females (95% CI: 4.11–10.22). The first systematic review and meta-analysis of this literature, conducted by Tian and Gao (2026) across 12 studies including over 42,000 ASD youth, confirmed a pooled hazard ratio of 3.64 for PP in ASD (95% CI: 1.42–9.34). These findings establish that ASD is associated with a substantially elevated population-level risk of precocious puberty, particularly in females, with effect sizes that are large and consistent across health systems and countries.

Neurobiological Mechanisms

The neurobiological mechanisms underlying atypical pubertal timing in ASD are not fully elucidated but converge on two primary pathways: adrenarche alterations and hypothalamic-pituitary-gonadal (HPG) axis dysregulation. Szakal and Crespi (2025) conducted a series of systematic reviews examining the hypothesis that alterations in adrenarche defined as the prepubertal phase involving onset of dehydroepiandrosterone (DHEA) and DHEAS secretion, typically beginning between ages 7 and 10 may be implicated in both the developmental expression of autistic traits and the timing of subsequent gonadarche. Adrenarche represents a neurologically significant hormonal transition linked to the development of complex ToM and social competencies during middle childhood; premature adrenarche (PA) has been associated with anxiety, depression, and emotional dysregulation in typically developing populations. The hypothesis that PA in ASD may disrupt the critical 7-to-10-year window during which second-order false belief understanding, faux pas recognition, and affective mentalising undergo

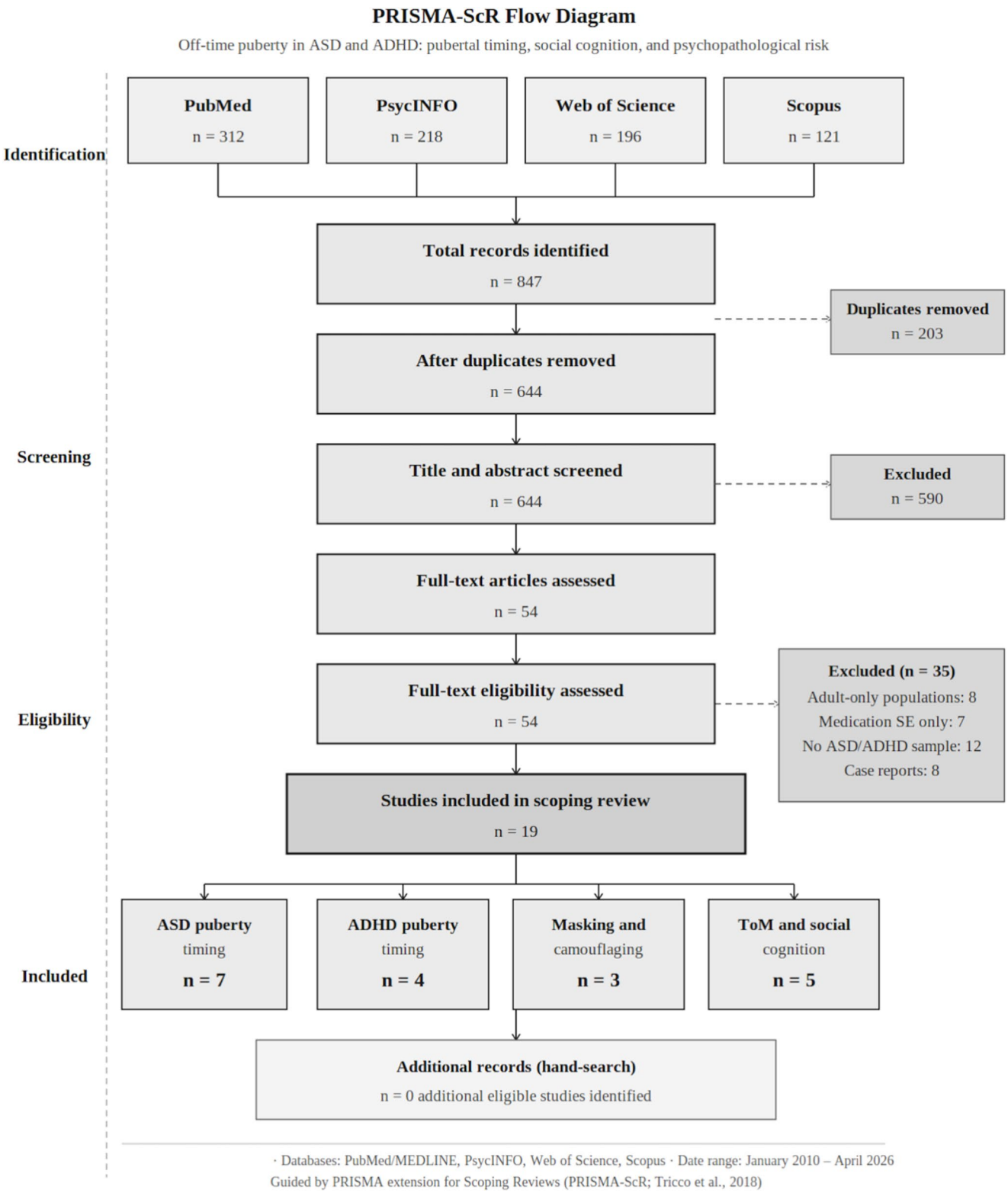


Fig. 1 PRISMA extension for scoping reviews (PRISMA-ScR) flow diagram illustrating the study selection process for the scoping review on off-time puberty in autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD)

Table 1 JBI methodological quality assessment of included primary studies

Study	Design	*N*	Puberty measure	Control group	Confounders controlled	Sex stratified	JBI quality	Overall
Corbett et al. (2020)	Cross-sectional	239	Tanner staging (physical exam)	Yes (TD)	Age, sex, IQ	Yes	7/8	High
Corbett et al. (2022)	Longitudinal (3-wave)	244	Tanner staging (physical exam)	Yes (TD)	Age, sex, IQ, race	Yes	8/8	High
Muscatello et al. (2022)	Longitudinal (1-wave)	244	Physical exam + salivary testosterone	Yes (TD)	Age, sex, puberty stage	Yes	7/8	High
Corbett et al. (2025)	Longitudinal (4-wave)	245	Physical exam & salivary hormones	Yes (TD)	Sex, puberty, diagnosis	Yes	8/8	High
Geier and Geier (2021)	Retrospective cohort	101,736	ICD-9 clinical diagnosis (CPP)	Yes (non-ASD)	Age, sex	Yes	5/8	Moderate
Tsai et al. (2024)	Nationwide cohort	22,208	ICD clinical diagnosis (CPP)	Yes (matched controls)	Age, sex, comorbidities	Yes	6/8	Moderate-high
Tian and Gao (2026)	Systematic review and meta-analysis	42,000	Multiple (clinical, Tanner, self-report)	Yes (TD)	PRISMA; heterogeneity assessed	Yes	8/8	High
Szakal and Crespi (2025)	Systematic review (series)	>3,900	Adrenarche markers (DHEA/DHEAS)	N/A (review)	Narrative synthesis	Partial	5/8	Moderate
Read et al. (2024)	Narrative review	>1,400	Multiple (clinical)	N/A (review)	Narrative synthesis	Yes	6/8	Moderate-high
Ostojic and Miller (2016)	Cross-sectional	253	Self-report (pubertal onset recalled)	No	Age, sex	Female only	4/8	Moderate
Rosenthal and Hinshaw (2024)	Longitudinal cohort	209	Tanner staging & menarche age	Yes (TD females)	Age, IQ, comorbidities	Female only	7/8	High
Osianlis et al. (2025)	Systematic review	>1.700	Multiple (self-report, hormonal)	Variable	PRISMA; narrative synthesis	Female only	6/8	Moderate-high
Mlodnicka et al. (2024)	Longitudinal cohort (ECHO)	2,818	APHV (peak height velocity)	Yes (non-ADHD)	Sex, age, ADHD severity	Yes	7/8	High
McKinney et al. (2024)	Cross-sectional (participatory)	119	Self-report (pubertal stage)	Yes (neurotypical girls)	Age, NDD diagnosis	Female only	5/8	Moderate
Bernardin et al. (2021)	Cross-sectional (online)	140	Not assessed	Yes (non-autistic)	Age, sex, diagnosis	Yes	5/8	Moderate
Khudiakova et al. (2025)	Systematic review	N/A	N/A (review)	N/A	Narrative synthesis	Yes	6/8	Moderate-high
Meinhardt-Injac et al. (2020)	Cross-sectional	267	N/A (ToM focus)	Yes (age-based)	Age, language, reasoning	Yes	6/8	Moderate-high
Morningstar and Burns (2025)	Longitudinal (ABCD)	9,648	PDS (parent-report)	Yes (within-sample)	Age, sex, site	Yes	7/8	High
Capuozzo et al. (2024)	Systematic review	>2,000	N/A (ToM/social cognition review)	Yes (controls)	PRISMA; ADHD diagnosis required	Yes	7/8	High

JBI quality scores are reported as number of criteria met out of 8. Overall quality: High = ≥ 7/8; Moderate-high = 5–6/8; Moderate = 3–4/8. *APHV* age at peak height velocity, *CPP* central precocious puberty, *PDS* Pubertal Development Scale. *TD* typically developing

their most rapid developmental progression carries substantial clinical significance, though direct longitudinal evidence linking adrenarche timing to ToM development in ASD remains to be established.

Read and colleagues (2024), reviewing the overlap between disorders of puberty and neurodevelopmental conditions, documented shared brain circuitry between the neuroendocrine systems controlling puberty onset and the social brain networks underlying ASD-associated features. The review reported that rates of central precocious puberty were highest in patients with ASD (16%), psychomotor delay (48%), and epilepsy (20%), suggesting a significant genetic and neurobiological component to the ASD-puberty association that may involve shared vulnerability in hypothalamic-limbic circuits governing both GnRH secretion and social motivation. These findings converge with evidence from genetic studies indicating shared polygenic architecture between autistic traits and earlier pubertal timing in females (Warrier et al., 2020, 2024).

Pubertal Timing in ADHD

Evidence regarding pubertal timing in ADHD, while less extensive than that for ASD, points to a pattern of earlier pubertal onset particularly in females, with evidence for sex-differentiated hormonal modulation of ADHD symptom expression across puberty. Ostojic and Miller (2016) reported a significant association between earlier self-reported menarche and greater ADHD severity in a sample of female university students (r = .21, p < .01), providing preliminary evidence for a female-specific puberty-ADHD relationship. Rosenthal and Hinshaw (2024), in an all-female longitudinal sample, confirmed earlier pubertal timing in ADHD relative to TD females using Tanner staging and menarche age, and demonstrated that earlier timing was associated with elevated internalizing symptoms including anxiety and depression at adolescent follow-up. In a large prospective cohort analysis using the ECHO consortium (Mlodnicka et al., 2024), pubertal timing as indexed by age at peak height velocity moderated the relationship between childhood ADHD symptoms and later psychopathology differentially by sex: earlier APHV in ADHD females predicted elevated conduct disorder symptoms, while later APHV in ADHD males was associated with depression and aggression at follow-up. Osianlis and colleagues (2025), in a systematic review spanning studies from 1980 to January 2025 (11 studies included), synthesised evidence that sex hormone fluctuations particularly estrogen decline modulate ADHD symptom severity in females through dopaminergic and noradrenergic pathways, with puberty representing the first major hormonal inflection point at which this sensitivity becomes clinically apparent. The review highlighted that ADHD in females is systematically underdiagnosed during childhood and adolescence, with the diagnostic gap widening precisely during puberty when hormone-driven symptom fluctuations may be misattributed to mood disorder or adjustment difficulties.

Masking, Camouflaging, and Psychopathological Risk at Pubertal Transition

Three included studies examined the mental health consequences of masking and camouflaging the deliberate suppression of autistic behaviours and effortful performance of social scripts at the pubertal transition in ASD and ADHD populations. McKinney and colleagues (2024), using participatory methods with neurodivergent girls (ASD, ADHD, and developmental coordination disorder) at the transition to adolescence, found that camouflaging predicted anxiety and depression symptoms, with the mental health gap between neurodivergent and neurotypical girls widening during the pubertal period. Bernardin and colleagues (2021), in a sample of 140 adolescents aged 13 to 18 years (62 non-autistic controls), demonstrated that camouflaging predicted internalizing symptoms ($\beta = 0.31$, p < .001) and externalizing symptoms after controlling for age, sex, and IQ, with internalizing symptoms mediating the camouflaging-externalizing relationship. Khudiakova and colleagues (2025), in a comprehensive systematic review of the camouflaging-mental health literature, noted that the association between camouflaging and mental health is inconsistent across cross-sectional studies and that no longitudinal studies have been published, highlighting a significant gap in understanding the developmental trajectory of camouflaging and its mental health consequences across the pubertal transition specifically.

Theory of Mind Development Across Adolescence: Normative and NDD-Specific Evidence

Three included studies examined ToM development across adolescence or its neurobiological correlates, providing normative context for understanding how ToM impairment in ASD and ADHD may interact with pubertal development. Meinhardt-Injac and colleagues (2020), in a cross-sectional study of 267 participants aged 11 to 25 years, demonstrated that social-perceptual ToM increases across adolescence with a significant age effect (η^2 = 0.08), and that females outperformed males in advanced social-cognitive ToM after controlling for language and reasoning a sex difference that becomes more pronounced precisely during the pubertal period. Morningstar and Burns (2025), leveraging ABCD study data (n = 9,648) across two longitudinal waves two years apart, demonstrated that pubertal maturation predicted

Table 2 Summary of included studies: authors, NDD group, country, sample size, diversity, design, puberty measure, key findings, and effect sizes

Study	NDD group	Country	*N*	Sample diversity	Design	Puberty measure	Key finding	Effect size
Corbett et al. (2020)	ASD	USA	239	82% White, 7.9% Black	Cross-sectional	Physical exam (Tanner)	ASD females: earlier puberty onset vs. TD (p=.018 for menarche)	Cohen d = 0.52 (females)
Corbett et al. (2022)	ASD	USA	244	83.3% White, 7.9% Black, 8.7% multiracial	Longitudinal (3-wave)	Physical exam (Tanner)	ASD: earlier onset both sexes; faster tempo in males	SMD reported by subgroup
Muscatello et al. (2022)	ASD	USA	244	Racially diverse	Longitudinal (cross-sectional wave)	Physical exam & salivary testosterone	ASD youth: significantly elevated testosterone vs. TD; sex × age interaction	Significant group effect (p<.001)
Corbett et al. (2025)	ASD	USA	245	Racially diverse	Longitudinal (4-wave, 10-17y)	Physical exam & salivary hormones (T and cortisol)	ASD males: testosterone mediated behavioral outcomes across puberty; HPA-HPG axis coupling	Mediation effects reported
Geier and Geier (2021)	ASD	USA	101,736	Florida Medicaid (economically diverse)	Retrospective cohort	ICD-9 clinical CPP diagnosis	PP incidence 3.15× higher in ASD; HR = 4.64	HR = 4.64 (95% CI: 3.11–6.93)
Tsai et al. (2024)	ASD	Taiwan	22,208	Taiwanese national cohort	Nationwide cohort	ICD clinical CPP diagnosis	ASD independent risk factor for PP; HR = 6.48 females	HR = 6.48 (95% CI: 4.11–10.22)
Tian and Gao (2026)	ASD	Multinational	42,017 (ASD)	Multiple countries	SR & meta-analysis (12 studies)	Multiple (clinical, Tanner, parent-report)	Pooled HR = 3.64 for PP in ASD; SMD for pubertal timing significant	Pooled HR = 3.64 (95% CI: 1.42–9.34)
Szakal and Crespi (2025)	ASD	Multinational	>3,900	N/A	Systematic review (series)	Adrenarche markers (DHEA/DHEAS)	Premature adrenarche linked to anxiety, depression, autistic trait expression; 7-10y critical window	N/A (narrative)
Read et al. (2024)	ASD and NDD	UK and Spain	>1,400	N/A	Narrative review	Clinical CPP diagnosis	ASD: 16% CPP comorbidity; shared HPG-social brain circuitry; ID × CPP documented	Descriptive
Ostojic and Miller (2016)	ADHD	Canada	253	Not reported	Cross-sectional	Self-report recalled menarche	Earlier menarche linked to greater ADHD severity in females	r=.21 (p<.01)
Rosenthal and Hinshaw (2024)	ADHD	USA	209	Predominantly White	Longitudinal	Tanner staging & menarche	Earlier pubertal timing in ADHD females; associated with internalizing symptoms	Significant group differences (p<.05)
Osianlis et al. (2025)	ADHD	Multinational	>1.700	Predominantly female adults	SR (PRISMA)	Multiple hormonal markers	Estrogen × dopamine: ADHD symptoms exacerbate at puberty and across menstrual cycle in females	Narrative synthesis
Mlodnicka et al. (2024)	ADHD	USA	2,818	Diverse (ECHO)	Longitudinal cohort	APHV (peak height velocity)	Earlier APHV in ADHD females; conduct disorder; later APHV in males; depression and conduct	Sex-moderated interaction effects
McKinney et al. (2024)	ASD and ADHD	UK	119	Girls only (UK)	Cross-sectional (participatory)	Self-report pubertal stage	Camouflaging predicted anxiety and depression at pubertal transition in neurodivergent girls	β coefficients reported
Bernardin et al. (2021)	ASD	USA	140	62 non-autistic controls	Cross-sectional	Not assessed	Camouflaging predicted internalizing and externalizing symptoms controlling for age, sex, IQ	β = 0.31 internalizing (p<.001)
Khudiakova et al. (2025)	ASD	Multinational	N/A	Mixed	SR	N/A	Camouflaging-mental health link inconsistent cross-sectionally; no longitudinal RCT; transactional IM framework proposed	Narrative
Meinhardt-Injac et al. (2020)	TD (ToM baseline)	Germany	267	Ages 11–25; mixed sex	Cross-sectional	N/A (ToM assessment)	Social-perceptual ToM increases through adolescence; gender difference in social-cognitive ToM	η^2=0.08 age effect

Table 2 (continued)

Study	NDD group	Country	*N*	Sample diversity	Design	Puberty measure	Key finding	Effect size
Morningstar and Burns (2025)	TD and varied	Canada	9,648 (ABCD)	ABCD study diversity	Longitudinal (2-wave)	PDS (parent-report)	Pubertal maturation predicts neural response changes to emotional faces (mentalizing regions: TPJ, amygdala)	β significant; salience and mentalizing networks
Capuozzo et al. (2024)	ADHD	Multinational	1,215 (22 studies)	Mixed	SR (PRISMA)	N/A (ToM review)	17/22 studies: ADHD shows significant deficits in ToM and EF vs. controls; affective ToM specifically impaired	Variable; most studies $p<.05$

ADHD attention-deficit/hyperactivity disorder, *APHV* age at peak height velocity, *ASD* autism spectrum disorder, *CPP* central precocious puberty, *ECHO* Environmental influences on Child Health Outcomes, *HR* hazard ratio, *IQ* intellectual quotient, *N/A* not applicable, *PDS* Pubertal Development Scale, *PP* precocious puberty, *SR* systematic review, *TD* typically developing, *ToM* theory of mind, *TPJ* temporoparietal junction

within-person changes in neural activation to emotional faces in mentalizing-relevant regions, specifically the temporoparietal junction and amygdala, after controlling for chronological age providing direct neural evidence that puberty shapes the brain systems underpinning ToM. Capuozzo and colleagues (2024), in a systematic review of 22 studies (1,215 ADHD participants), confirmed that 17 of 22 studies found significant deficits in ToM and executive function in ADHD relative to controls, with affective ToM showing the most consistent impairment profile.

A summary of all included studies, including sample characteristics, design, puberty measurement, and key findings, is presented in Table 2.

Discussion

The evidence synthesised in this review supports a view of off-time puberty in ASD and ADHD as a developmentally amplifying risk factor rather than an additive one. The consistency of the ASD-early puberty association across physical examination-based longitudinal studies, large population cohorts, and meta-analytic synthesis (pooled HR = 3.64 for precocious puberty) establishes altered pubertal timing in ASD as a replicable and clinically significant finding. The ADHD-puberty relationship, while less well-characterised, is supported by converging evidence from longitudinal studies and systematic review, with a clear sex-differentiated profile: earlier puberty in females, with hormonal modulation of symptom expression through estrogen-dopamine pathways. Neither of these literatures, however, has been examined in conjunction with the social-cognitive profile of affected youth specifically, with the ToM capacity that determines how effectively the child can navigate the increasingly complex social demands that pubertal development introduces.

Drawing on the convergent evidence reviewed, we propose a dual-vulnerability framework with three specific mechanisms. The first, demand-capacity mismatch amplification, posits that off-time puberty particularly early pubertal timing accelerates the rate at which social demands escalate, producing a wider and faster-growing gap between the social-cognitive demands of the peer environment and the child's mentalising capacity. An early-maturing autistic girl faces the social demands of mid-adolescence with the mentalising resources of a child whose social cognition is both impaired and developing more slowly than TD peers. The second mechanism, masking and camouflaging under social load, recognises that autistic females and, to a lesser extent, females with ADHD engage in compensatory masking that is cognitively demanding, emotionally costly, and associated with elevated anxiety, depression, and delayed

diagnosis. Puberty substantially increases the social load under which masking must be sustained: the social environment becomes more complex precisely when hormonal changes are simultaneously increasing emotional reactivity. Earlier pubertal timing compresses the developmental window during which masking strategies can be consolidated, potentially increasing their fragility under elevated social load. The third mechanism, physical-cognitive developmental asynchrony, identifies a vulnerability unique to NDD populations: physical maturation outpaces social-cognitive developmental level, generating social expectations from peers and adults that substantially exceed the child's actual mentalising capacity creating specific vulnerability to social exploitation, peer victimisation, and social isolation.

Critically, the review identifies a significant primary evidence gap: no published study has directly examined whether baseline ToM capacity moderates the association between pubertal timing and psychopathological outcomes in ASD or ADHD populations. This represents the most important research priority arising from the synthesis. Future longitudinal studies should: (1) assess baseline ToM using performance-based measures (false belief tasks, Faux Pas Recognition, CANTAB ERT) at puberty onset and examine whether ToM profile predicts steeper mental health deterioration in off-time developers; (2) include assessment of puberty using gold-standard physical examination-based Tanner staging alongside hormonal biomarkers rather than parent or self-report alone; (3) include adequate representation of female participants across ASD and ADHD, given the sex-differentiated nature of both the puberty effects and the masking mechanisms; and (4) follow participants longitudinally from pre-puberty through mid-adolescence to capture the temporal dynamics of the puberty-ToM-psychopathology relationship.

Clinically, the framework argues for several adjustments to practice. First, assessment protocols for NDD youth aged 8 to 12 years should systematically document pubertal stage and timing, identifying off-time developers as a priority subgroup for enhanced mental health monitoring. Second, social cognition and social skills intervention content for NDD youth should be calibrated to pubertal stage as well as chronological age: an autistic girl who is experiencing early puberty at age 10 faces social demands analogous to those faced by a TD 13-year-old. Third, diagnostic re-evaluation is warranted for ADHD females presenting in early adolescence with internalising symptom escalation coinciding with pubertal onset, as this pattern may reflect hormone-modulated ADHD rather than primary mood disorder. Fourth, family and educator psychoeducation should address the physical-cognitive developmental asynchrony in ASD and ADHD, helping caregivers calibrate social expectations to cognitive rather than physical developmental level.

Limitations

Several limitations of this review should be noted. First, the scoping methodology does not permit meta-analytic pooling of primary studies; effect size estimates reported are derived from individual studies rather than synthesised. Second, the evidence base for ADHD is substantially smaller and less methodologically rigorous than that for ASD, with no Tanner staging-based longitudinal study in an ADHD-diagnosed sample identified for females. Third, the literature on DLD and ID was excluded to permit depth of synthesis in the populations with the strongest evidence base; this limits its generalisability to the broader NDD spectrum. Fourth, the overwhelming majority of the reviewed literature comes from high-income English-speaking countries, limiting generalisability to other cultural and socioeconomic contexts. Fifth, no primary study directly testing the key hypothesis of this review that ToM moderates the puberty-psychopathology relationship in NDD populations was identified, rendering the dual-vulnerability framework necessarily theoretical at this stage.

Conclusion

Off-time pubertal development and pre-existing ToM impairment represent two well-documented but separately studied risk factors for adolescent psychopathology in ASD and ADHD. This scoping review is the first to systematically map the evidence at their intersection, identifying a substantial and consistent literature on altered pubertal timing in ASD, emerging evidence in ADHD with a clear sex-differentiated profile, and converging neurobiological mechanisms linking puberty to social-cognitive brain development. The absence of any primary study directly examining ToM as a moderator of the puberty-psychopathology relationship in NDD populations represents the most critical gap in this literature. The dual-vulnerability framework proposed here integrating demand-capacity mismatch amplification, masking under social load, and physical-cognitive developmental asynchrony as specific mechanisms is intended to provide the theoretical scaffolding for the next generation of longitudinal and intervention research in this clinically important and underserved population.

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Declarations

Ethics Approval Not applicable. This scoping review does not involve original data collection or human subjects research.

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